Integration and Visualization of Meteorological Data in Digital Twins: A Framework for Real-time Weather Simulation Using AnywhereXR Platform

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Background and Intro

The increasing frequency and intensity of extreme weather events demand sophisticated visualization capabilities within digital twin environments. Current meteorological data integration faces significant challenges in achieving real-time, high-fidelity weather representation while maintaining computational efficiency.

Digital twin technologies have evolved beyond industrial applications to enable comprehensive environmental monitoring. However, the integration of meteorological phenomena presents unique challenges in data processing and visualization. The HARMONIE-AROME numerical weather prediction model provides high-resolution meteorological data (2km × 2km grid) but requires sophisticated interpolation strategies for seamless visualization.

Objective and Research questions

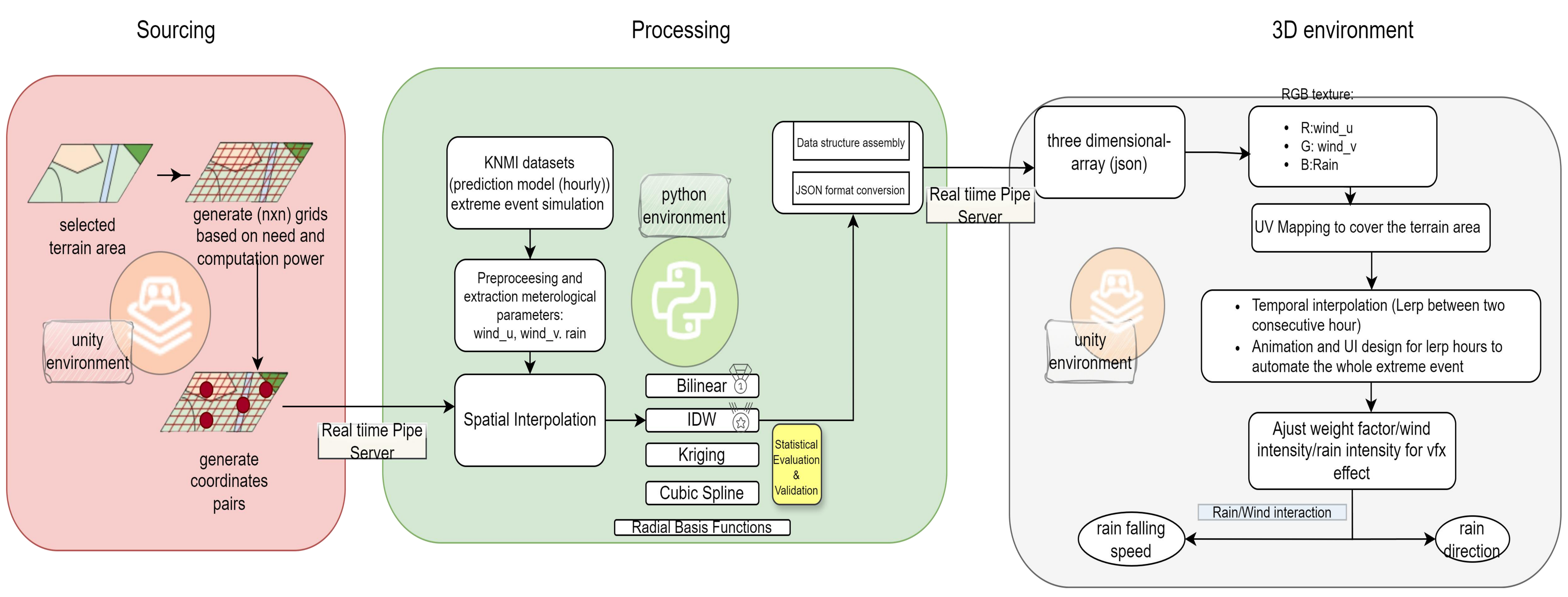
This study addresses three critical challenges:

RQ1: How can an effective prototype framework be developed to facilitate bidirectional data communication between Python-based meteorological processing and Unity-based visualization systems?

RQ2: Which spatial interpolation methodologies optimize the balance between computational efficiency and physical accuracy in meteorological parameter reconstruction?

RQ3: How can temporal continuity be achieved in three-dimensional weather visualization while maintaining physical authenticity of meteorological phenomena?

Methods



Sourcing

- Terrain area selection with 64×64 adaptive sampling grid
- Real-time coordinate generation for interpolation points

Processing

- KNMI HARMONIE-AROME data integration for extreme weather
- Five interpolation methods implementation
- Statistical validation and optimization
- Real-time data structure assembly

3D Environment

- RGB texture encoding for wind and precipitation
- Temporal interpolation and continuous animation
- Dynamic VFX parameter optimization

Results

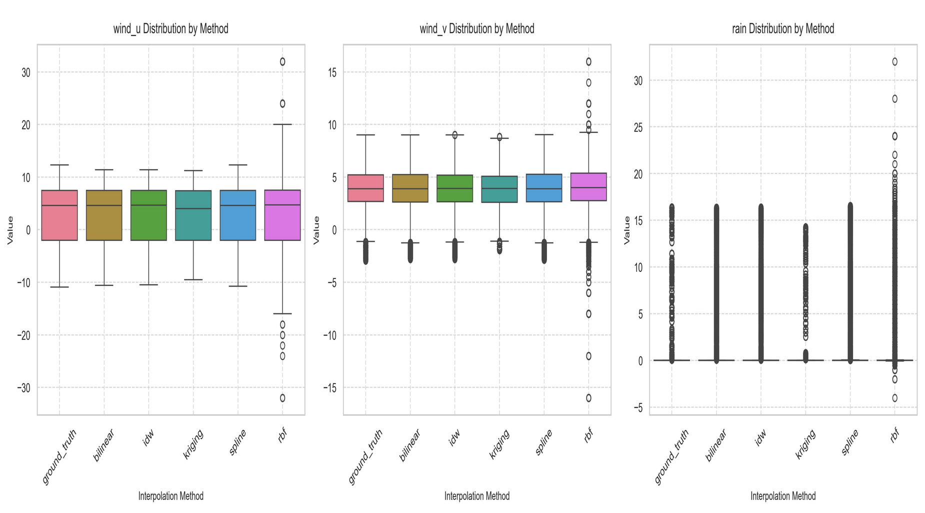


Figure1 Distribution of Interpolated Values by 5 various methods

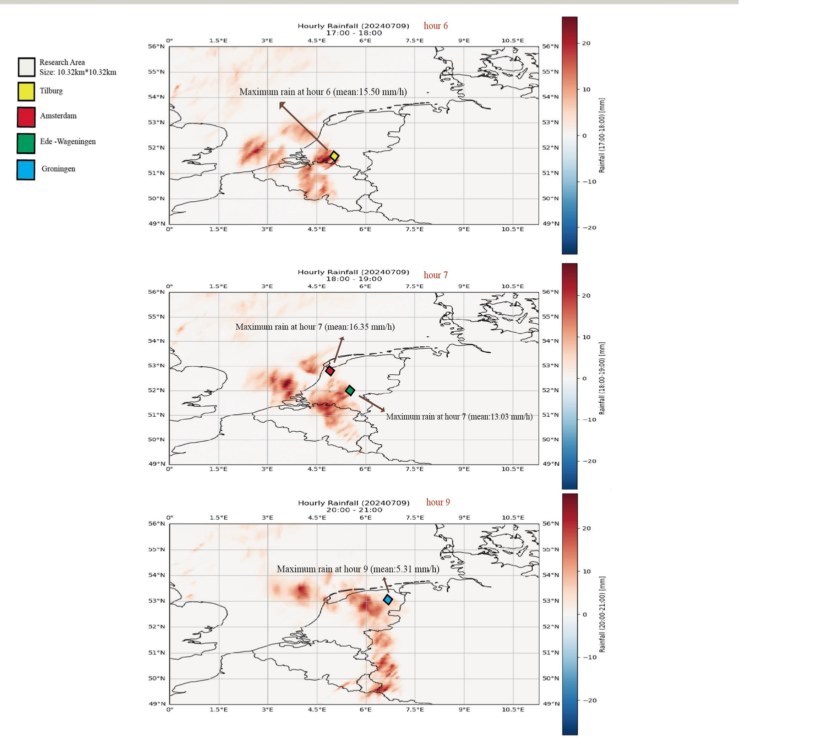


Figure2 Time series analysis of rain parameter in 4 test locations

Table1 Comprehensive Method Performance Analysis (one way ANOVA)

Method	Parameter	F-statistic	p-value	Best Location (ME)	Worst Location (ME)
Bilinear	WindU	517.84	<0.001	Groningen (0.0836)	Amsterdam (0.1260)
	WindV	1452.03	<0.001	Groningen (0.0744)	Amsterdam (0.1404)
	Rain	1436.20	<0.001	Groningen (0.0179)	Amsterdam (0.0759)
IDW	WindU	317.91	<0.001	Groningen (0.0856)	Amsterdam (0.1211)
	WindV	1062.37	<0.001	Groningen (0.0789)	Amsterdam (0.1387)
	Rain	1154.13	<0.001	Groningen (0.0185)	Amsterdam (0.0692)

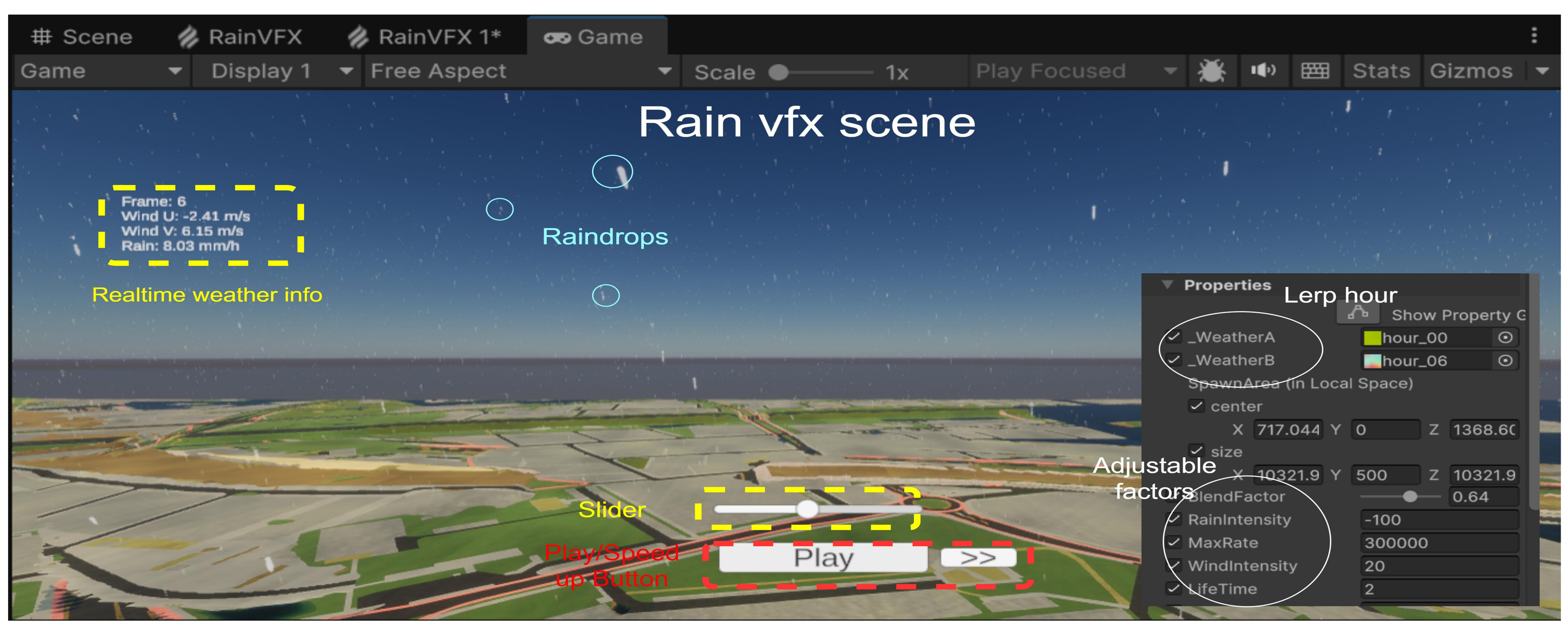


Figure4 Rain vfx scene implementation in unity

Conclusion

- **Bidirectional Communication Framework Achievement**
  - Implementation of a structured named pipe architecture enabling atomic data transmission
  - Achievement of reliable coordination between Python processing and Unity visualization through message-based protocols
- **Spatial Interpolation Performance**
  - Bilinear interpolation achieves optimal performance for wind fields ($R^2 > 0.99$)
  - IDW demonstrates superior accuracy in precipitation reconstruction (RMSE < 0.27 mm/h)
  - Significant geographical sensitivity ($p < 0.001$) across all methods
- **Visualization Integration**
  - RGB texture encoding of meteorological parameters
  - Smooth temporal transitions between forecast hours by lerp interpolation between hours
  - Dynamic VFX parameter adjustment for realistic weather representation

Discussion

- How can spatial resolution be optimized dynamically?
- What underlying geographical factors influence interpolation performance?
- Can real-time automation be achieved across the entire visualization pipeline?

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